



Adaptive Sampling for Sparse Sensor Networks

A field study on the Upper Rhine

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September 19, 2026 · ICSE Workshop, Mainz

Outline

1 Motivation

Where static sampling breaks down in the field

2 Method

A variance-driven scheduler with a submodular guarantee

3 Results

Eleven weeks on the Upper Rhine, 18 nodes

01

Motivation

Motivation

Method

Results

Why sparse networks fail

Three observations from two deployments

- Node dropout is bursty, not uniform.
- Static sampling wastes 40 % of the energy budget.
- Reconstruction error grows superlinearly below 12 active nodes.

Takeaway

Sampling rate should follow measured field variance, not the clock.

The gap in the literature

Placement is treated as a one-off design decision, although the field it samples is non-stationary on the scale of days.

Krause & Guestrin, JMLR 2008

02

Method

Motivation

Method

Results

Problem statement

We minimise the expected reconstruction error under an energy constraint:

$$\min_{\pi} \mathbb{E} \left[\|x - \hat{x}_{\pi}\|_2^2 \right] \quad \text{s.t.} \quad \sum_t c(\pi_t) \leq E_{\max}. \quad (1)$$

Theorem 1 (Convex relaxation)

For submodular $c(\cdot)$ the relaxed problem admits a $(1 - 1/e)$ approximation.

Algorithm

Variance-driven scheduler

```
1  $S \leftarrow \emptyset$ 
2 while  $\text{budget}(S) < E_{\max}$  do
3    $v^* \leftarrow \arg \max_{v \notin S} \Delta(v \mid S)$ 
4    $S \leftarrow S \cup \{v^*\}$ 
5 end while
6 return  $S$ 
```


Reference implementation

```
1 def schedule(nodes, budget):
2     active = []
3     while cost(active) < budget:
4         best = max(nodes, key=lambda v: gain(v, active))
5         active.append(best)    # greedy, submodular gain
6     return active
```

03

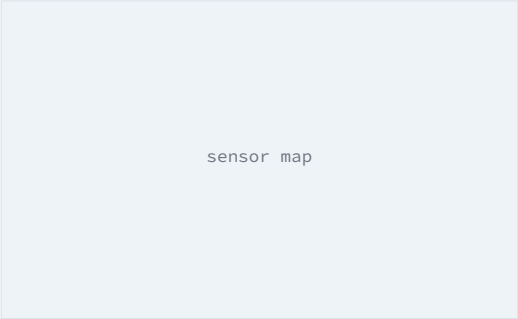
Results

Motivation

Method

Results

Field data



sensor map

- 18 nodes, 11 weeks
- 2.1 M samples
- 3 dropout events

Adaptive scheduling kept coverage above 92 % throughout.

Deployment area

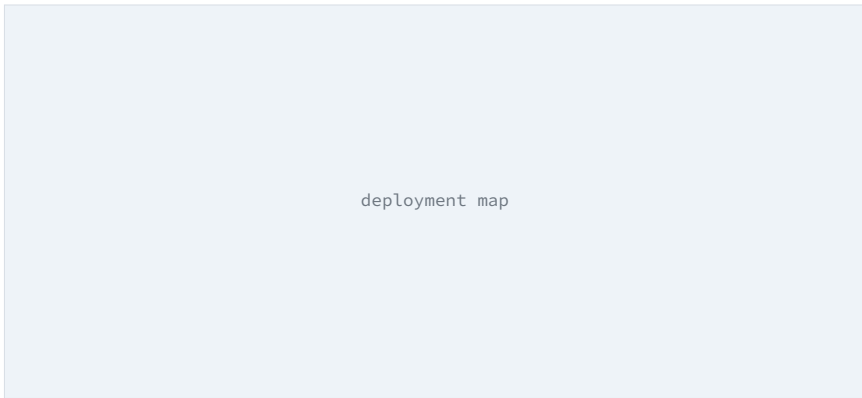
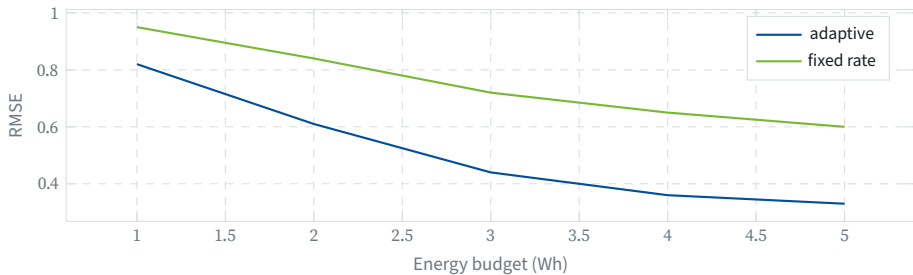


Fig. 1: Eighteen nodes along 11 km of the Upper Rhine; green nodes were rescheduled at least once.

Error versus energy budget



Headline result

39 %

lower reconstruction error than fixed-rate sampling at the same
energy budget

Comparison

Method	RMSE	Energy	Uptime
Fixed rate	0.72	3.0	88 %
Round robin	0.58	3.0	90 %
Adaptive (ours)	0.44	3.0	92 %

Averaged over 11 weeks; lower is better.

References

- A. Krause and C. Guestrin. *Near-optimal sensor placements*. JMLR, 2008.
- L. Zhu et al. *Energy-aware scheduling in environmental networks*. Sensors, 2021.

Thank you

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TAKEAWAYS

- Sampling should follow field variance, not the clock.
- 39 % lower error at an unchanged energy budget.
- Code and the 11-week dataset are public.



Backup

Sensitivity analysis

Slides after \thbbackup hold material for the questions after the talk.